

The 3D Viewer — and the Data It Captures

A competitive study of how the best should-cost tools turn CAD into cost automatically — and a build plan to make our 3D viewer feel like real CAD software.

Market study

Gap analysis

4-phase roadmap

Bigger, CAD-grade viewer

EXECUTIVE SUMMARY

What we found — and the bet

1 The pattern is universal

Every leader runs automatic feature recognition on the CAD B-rep, maps features to cost drivers, and lets a human CORRECT — never re-type — the geometry.

2 We are already strong where it is hard

Our OCCT engine already measures holes, depths, walls, draft, setups and feeds them to cost. Our real gaps are the VIEWER and showing what we already know.

3 The "feel" is on-model overlays

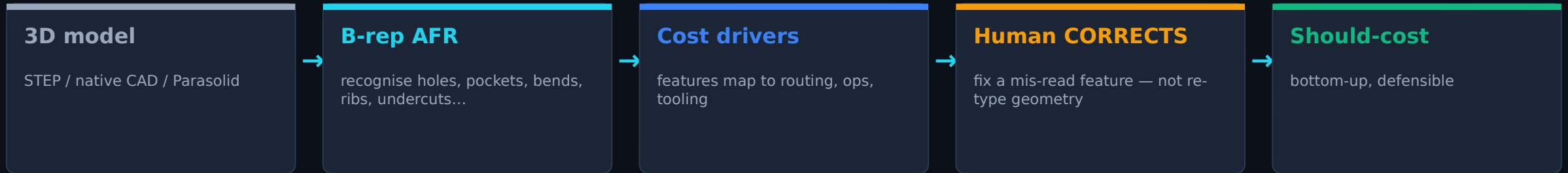
What makes a viewer feel like CAD software is DFM painted ON the model — thickness & draft heatmaps, section, a feature tree — plus room to breathe (a big canvas).

4 The build is bounded & staged

A 4-phase plan: quick wins first (bigger viewer + paint the data we already compute), then CAD-grade tools, then deeper capture (PMI/tolerances), then best-in-class AFR.

THE PATTERN · VERIFIED

How the best tools turn CAD into cost



aPriori — Geometric Cost Drivers

Inspects the raw B-rep — independent of how the native CAD built the feature — to extract named "Geometric Cost Drivers", then builds cost bottom-up. Because it costs from STEP / Parasolid (history-free formats), the recognition is provably topology-based, not feature-tree replay. GD&T can be added onto drivers to steer the process route.

"Analyzes the 3D model, breaks it down into individual geometric cost drivers, and builds a cost from the bottom up." — aPriori / DEVELOP3D

Siemens NX Feature2Cost

Commodity-specific modules (Molding, Stamping) automatically analyse the geometry file and recognise part dimensions, ribs, undercuts, bends and progressive-die features, turn them into cost drivers, allow manual correction, and pass them "seamlessly" into Teamcenter Product Cost Management / Tool Costing.

"Recognizes important geometrical information like general part dimensions, ribs, undercuts, bends... and drives them into actual cost drivers." — Siemens Teamcenter blog

THE LANDSCAPE

Competitor capability matrix

Tool	B-rep feature recognition	On-model DFM overlays	PMI / GD&T → cost	CAD-software viewer feel	Auto → cost (no re-entry)	Web / instant
aPriori	●	●	●	●	●	●
Siemens NX F2C	●	●	●	●	●	○
3D-Tool (viewer)	●	●	○	●	○	●
Werk24 (drawings)	●	○	●	○	●	●
MTI Costimator 3DFX	●	○	○	●	●	○
CostVision — today	●	○	○	●	●	●
CostVision — planned	●	●	●	●	●	●

● Full
● Partial
○ Limited / none

Sources: aPriori (DEVELOP3D / product), Siemens Teamcenter blog, 3D-Tool, Werk24 docs, MTI Systems — all verified. Marks are our read of published capability.

THE EXPERIENCE

What makes a viewer "feel like CAD software"

It is not prettier rendering — it is analysis painted ON the model, and room to work. 3D-Tool is the gold standard: four heatmap analyses shown directly on the part.

Wall-thickness heatmap

Thin/thick zones coloured on the model (single-ray & fitted-ball methods) — instantly shows mould/casting risk and cost drivers.

Draft & undercut analysis

Draft angle shaded per face; areas that "cannot be demolded" flagged as undercuts → slide units → tooling cost.

Section & measurement

Live cut planes, true measure, model comparison and collision — the everyday CAD verbs users expect.

Feature / model tree

A persistent list of recognised features; click a row → the face lights up. Two-way link between the number and the geometry.

PMI / GD&T on the model

Tolerances and annotations shown in 3D, tied to the faces that carry the cost of holding them.

A big, fluid canvas

Fullscreen / large view, an orientation cube, smooth orbit — the difference between "a widget" and "a tool".

OUR BASELINE · FROM A CODE AUDIT

Where CostVision stands today — honestly

✓ Already strong

- OCCT B-rep engine: volume, bbox, area, weights, face/edge classes
- Real feature table — holes & bosses with Ø, depth, through/blind
- Ray-cast wall thickness, draft/undercut analysis, setup-count
- Sheet-metal bend detection; parametric tooling-cost estimates
- Strong AUTO → cost: geometry auto-fills the cost form, no re-typing
- Per-face B-rep intelligence on click (type, Ø, hole-vs-boss, area)
- Distance / 3-pt circle / angle measure, section plane, CSV export
- Runs in the browser — instant, no desktop install

△ The real gaps

- No bigger / fullscreen viewer — fixed ~560px panel
- We COMPUTE thickness, draft, setups — but never paint them on the model
- No feature/model tree, no orientation cube, no exploded view
- Mesh-only render; STEP is tessellated server-side (no live B-rep in browser)
- No PMI / GD&T / tolerance capture from STEP AP242
- Feature recognition stops at holes/bosses — no fillet, chamfer, slot, thread specs
- STL uploads are feature-blind (holes/draft/setups all zero)
- Report feature tables not click-linked back to the 3D model



WHERE THE LEVERAGE IS

The strategic insight

**“We already measure it —
we just don’t show it.”**

Wall thickness, draft, undercuts, setups and manufacturability are all computed server-side today and thrown away visually. Painting them on the model is the single highest ratio of perceived value to engineering effort in this whole plan.

**“Capture more —
so users type less.”**

Every feature we recognise from the CAD (fillets, chamfers, threads, and tolerances from AP242) is one less field a human fills in — and one less place the estimate can drift. More capture = faster, more defensible, more automatic.

Design principle we keep: the viewer is not decoration — it is an INPUT-CAPTURE surface. Every pixel earns its place by either capturing a cost driver or letting a human verify one. Glass-box, always: the AI reads geometry, it never sets a price in secret.

HOW · VERIFIED & BUILDABLE

The technology we would build on

occt-import-js (OCCT → WASM)

Import STEP/IGES/BREP in the browser. Its JSON exposes a "brep_faces" map tying mesh triangles back to original B-rep faces — exactly what enables per-face selection & highlighting linked to cost, live, with no server round-trip.

OCCT XDE — AP242 PMI

OpenCascade's XDE / STEPControl reads PMI (GD&T) from STEP AP242: dimensions, tolerances, datums. This is the path to reading tolerances straight from the 3D file — not a separate drawing.

Native B-rep AFR (deep learning)

UV-Net, BRepGAT, BrepMFR, AAGNet encode the B-rep as a face-adjacency graph and segment machining features per face — 98-99%+ on public benchmarks — WITHOUT lossy mesh/voxel conversion.

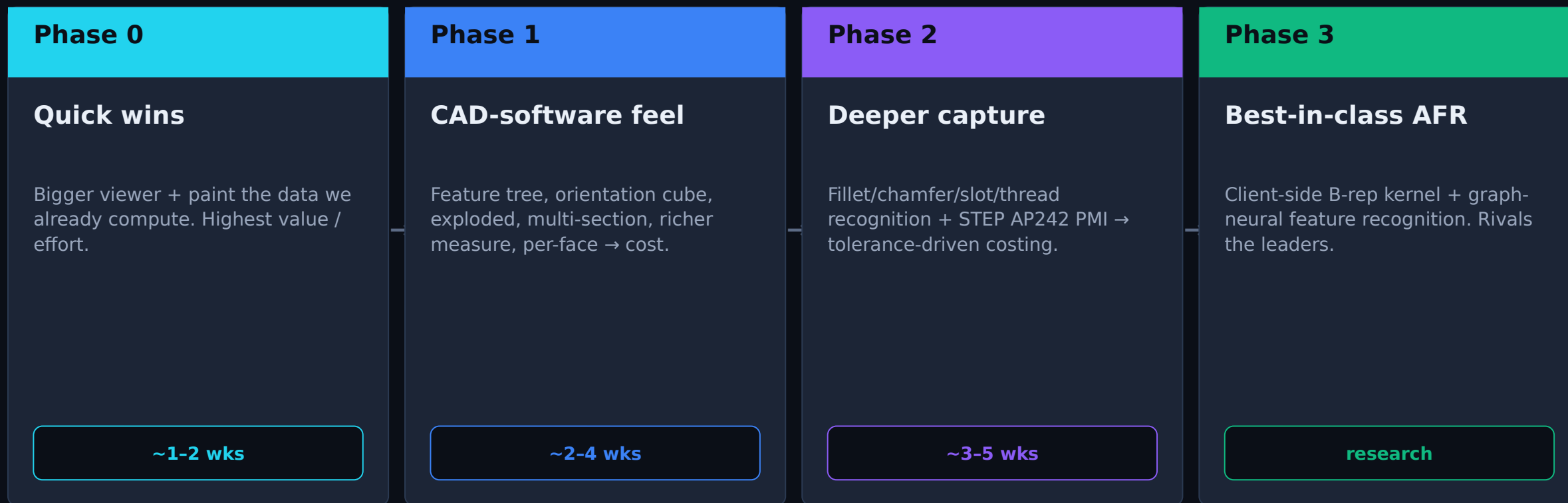
Heatmap methods (well-trodden)

Wall thickness = single-ray + fitted-ball (both textbook). Draft = face-normal vs pull direction. We already compute the numbers; rendering them is a shader/vertex-colour job.

Honest caveat: those 98-99% AFR figures are per-face on SYNTHETIC datasets; vendor "seamless / 95%+" claims are marketing. Real STEP is messier — so we keep the human-in-the-loop correction step, always.

ROADMAP

The build plan — four phases



Ship each phase on its own — value lands early. The two things you asked for (a bigger viewer, a CAD-software feel) are Phase 0 and Phase 1.

ROADMAP · ~1-2 WEEKS · HIGHEST LEVERAGE

Phase 0 — Quick wins

Everything here is high-value and low-risk — three of the five are just SHOWING data the engine already produces.

Bigger / fullscreen viewer

Maximise button + Fullscreen API + a large "focus" mode and drag-to-resize.
Directly the ask: room to work.

→ Instantly a serious tool; better demos & inspection

Wall-thickness heatmap

Render the thickness we ALREADY compute as a colour map on the model, with a legend.

→ Thin-wall cost/mould risk visible at a glance

Draft / undercut overlay

Shade draft angle per face; flag undercuts (data already computed) → tooling drivers.

→ DFM at concept; explains die/slide cost

Persistent feature tree

Turn the existing feature table into a docked tree; click a row → highlight the face.

→ Two-way link between the number and the geometry

Two-way report ↔ model link

Click a machined-feature report row → the face lights up in 3D, and back.

→ Trust: every cost line traces to a face

ROADMAP · ~2-4 WEEKS · YOUR 2ND ASK

Phase 1 — The full CAD-software feel

Orientation cube + smooth views

A draggable nav cube and named-view snapping — the universal "this is CAD" signal.

Feature / model tree

A real docked tree of bodies, faces & recognised features with visibility toggles.

Exploded view

Separate the solids of an assembly along an axis — inspect and cost per component.

Multi-section + cap shading

More than one cut plane, with solid caps — proper sectioning, not a single slider.

Richer measurement

True face radius, edge length, face-to-face parallel distance, coordinate read-out.

Per-face → cost linking

Click a face → see the operation and £ it drives. The viewer becomes the cost story.

The payoff: the viewer stops being "a preview" and becomes the place engineers inspect, verify and understand the cost — the exact thing that impressed you about the competitor.

Phase 2 — Deeper automatic capture

More recognised features → cost

- Fillets & chamfers — count, size → finishing / deburr ops
- Slots & pockets — proper recognition (today only holes/bosses)
- Ribs & bosses on mouldings → tool complexity
- Thread specs — real pitch/class, not a boolean guess
- Machined-vs-as-cast face classification → what to machine
- Better STL path — recover features so STL isn't cost-blind

Tolerance-driven costing (the differentiator)

Read PMI / GD&T straight from STEP AP242 (via OCCT XDE) — the tolerances that actually drive cost — and map them to process & inspection:

Tight bore tolerance

→ ream / hone / grind added

Fine surface finish (Ra)

→ super-finish / polish op

Flatness / position GD&T

→ inspection + fixturing cost

Loose general tol.

→ cheaper route, faster cycle

ROADMAP · RESEARCH TRACK

Phase 3 — Best-in-class recognition

Client-side B-rep kernel (occt-import-js)

Open STEP/IGES directly in the browser and hold the real B-rep, not a server-tessellated mesh. Unlocks true face-radius measurement, instant load, and per-face picking with zero round-trip.

Benefit: the viewer stops depending on a server call to feel alive — and every face is a first-class, clickable, costable object.

Graph-neural feature recognition (AFR)

Add a native B-rep AFR model (UV-Net / BRepGAT family) that segments machining features per face at published 98–99% on benchmarks — recognising the compound features our geometric heuristics miss.

Benefit: recall and precision that rival aPriori & Siemens — while staying glass-box, because every prediction is a face a human can verify.

Why this is a "later" phase, not a "never" phase

Highest effort and the benchmark accuracy is on synthetic data — so it earns its place only after Phases 0–2 have already closed the visible gap. But it is the ceiling: it is what would let us say "our recognition matches the market leaders," on our own secure, learning, glass-box platform.

WHAT YOU ASKED FOR, CONCRETELY

Your two asks — specced

1 · Make the 3D viewer bigger

- Maximise button in the toolbar → full-window overlay
- Browser Fullscreen API (true full-screen, F11-style)
- Large "focus" mode: viewer takes the whole workspace
- Drag-to-resize handle on the panel; remembers your size
- Toolbar & panels reflow so nothing crowds the model
- Escape / click-out returns to the costing layout

2 · Full CAD-software feel

- Orientation cube + named-view snapping
- Docked feature / model tree (click row → highlight face)
- On-model heatmaps: wall thickness, draft, undercut
- Multi-section planes with cap shading; exploded view
- Richer measure: face radius, edge length, face-to-face
- Per-face → cost: click a face, see its op and £

WHY IT IS WORTH IT

Benefits, and the recommendation

More auto-capture

Every feature & tolerance read from CAD is a field nobody types — faster quotes, fewer errors.

Fewer estimate drifts

Capturing the driver instead of typing it removes a place the number can go wrong.

DFM at concept

On-model thickness/draft/undercut catches cost problems while a change is still free.

Demo & sales credibility

A big, CAD-grade viewer is what wins the room — the exact reaction you had to the rival.

Tolerance-correct cost

Reading GD&T from the model prices precision parts right — a real differentiator.

Still glass-box & ours

Every captured value is a face a human can verify; secure, on-prem, self-learning.

Recommendation

Approve Phase 0 now — the bigger viewer plus painting the data we already compute, ~1-2 weeks. It delivers both of your asks fastest and de-risks the rest. Then take Phases 1-2 as a fast-follow. Phase 3 (graph-neural AFR) stays a research track we green-light once the visible gap is closed.